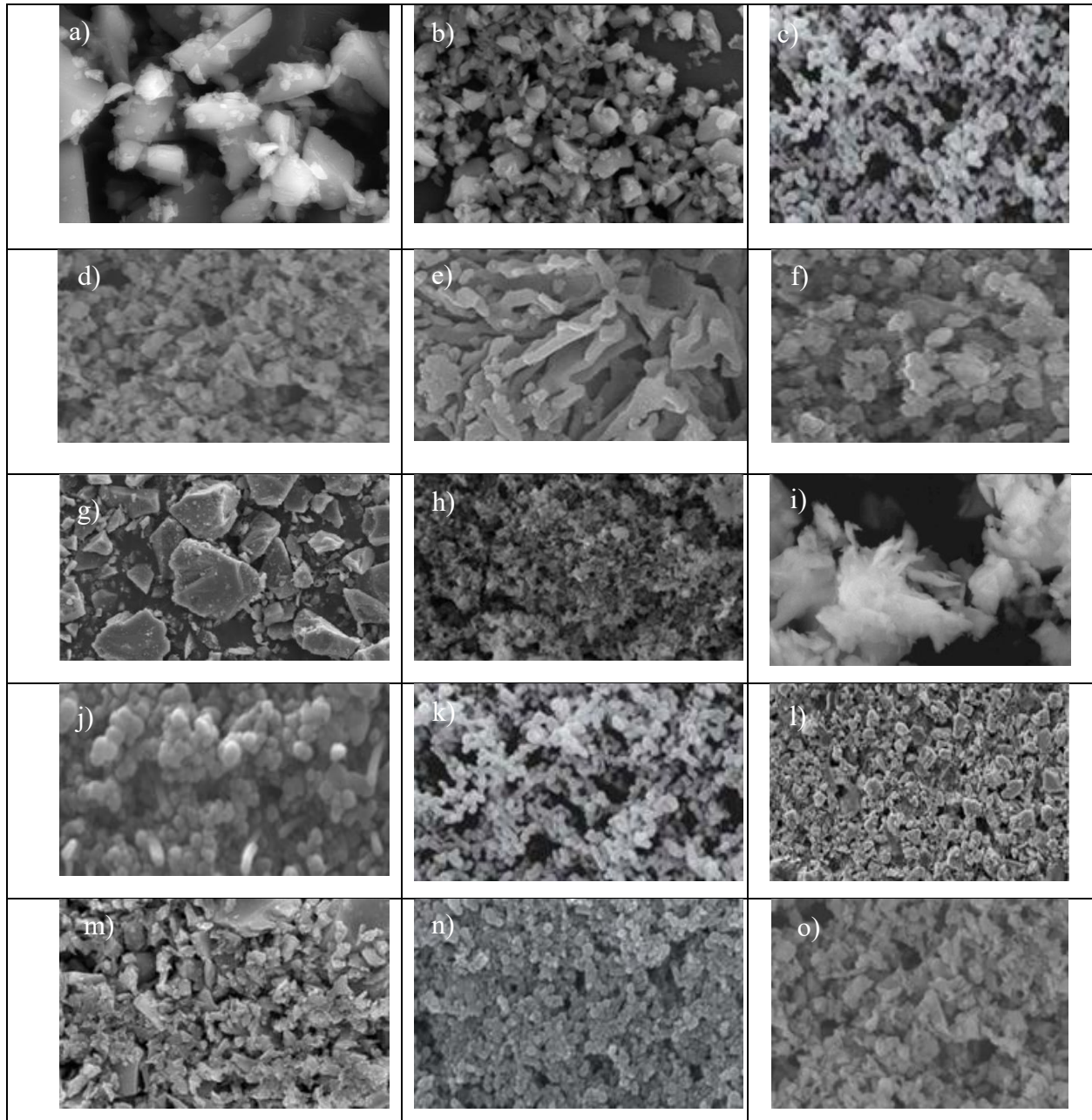
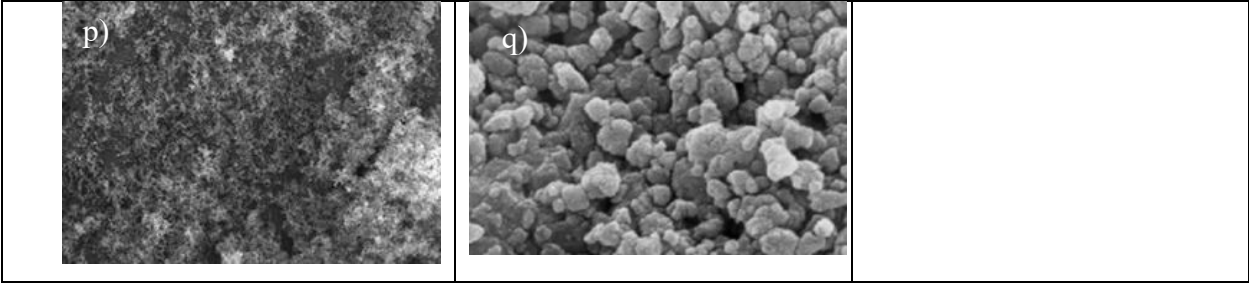


DATASET CITRA SEM SILIKON

Devinka Nursiva Rizantti

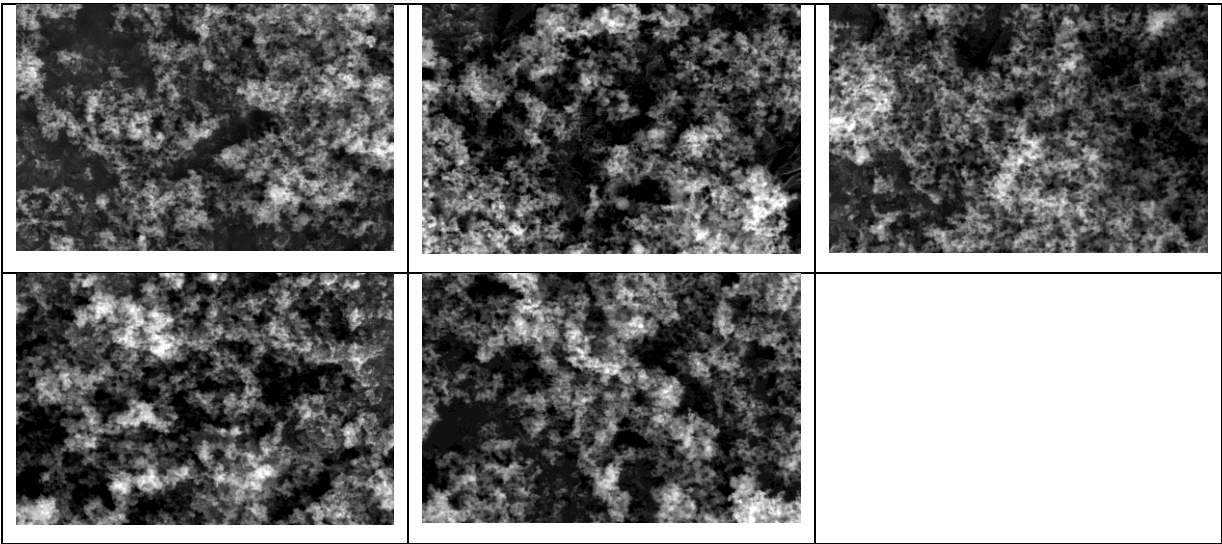
Departement of Physics, Universitas Airlangga 2026





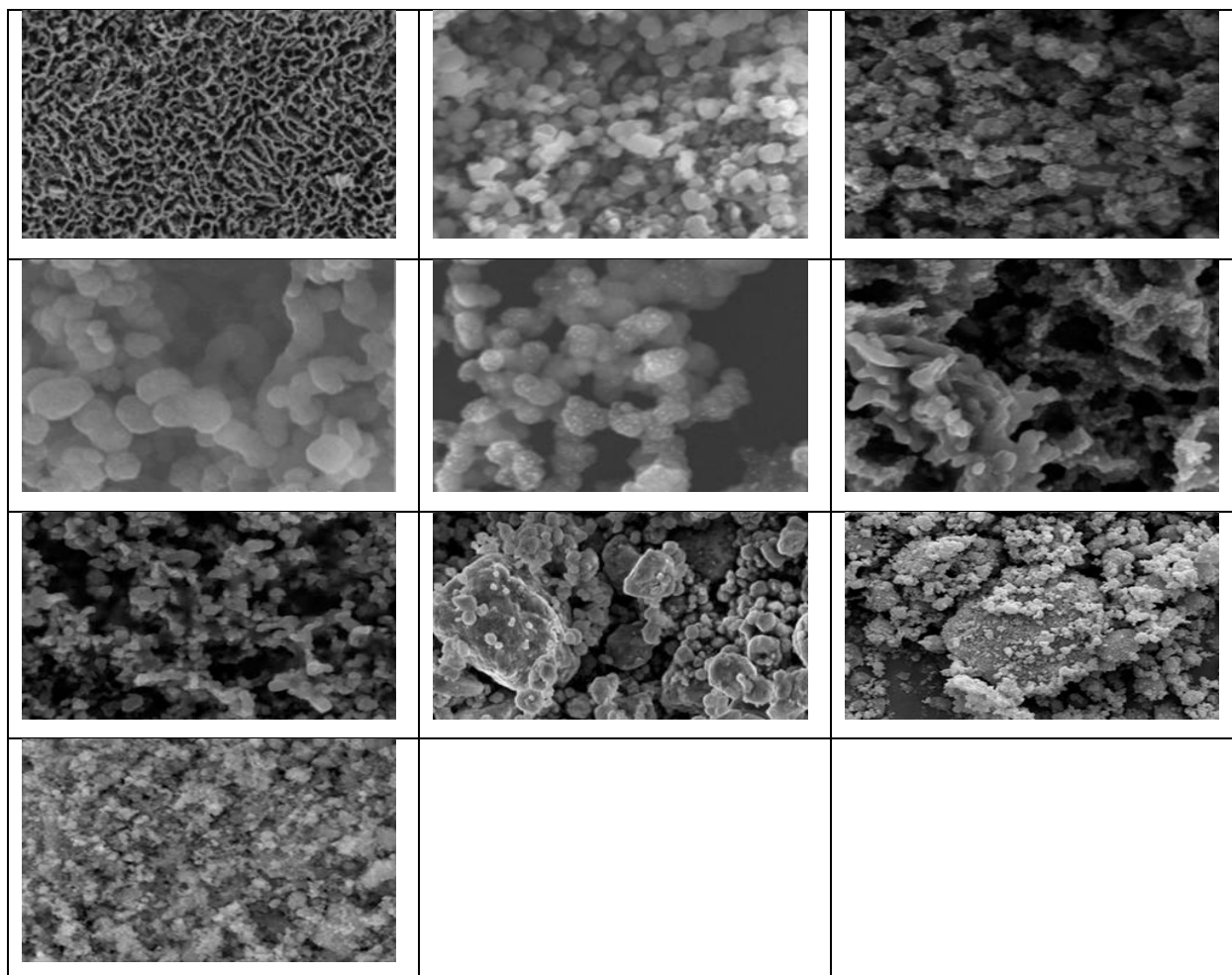
Lampiran 5. 1 Sampel Citra Silikon Murni a) data primer penelitian Silikon murni, (b)–(q) data sekunder dari beberapa jurnal.

Sumber : Data primer penelitian dan data sekunder yang diadaptasi dari (Khan et al., 2024), (Protopapa et al., 2025), (Xiao Zhongliang et al., 2024), (Jiang & Li, 2022a), (Wagner et al., 2019), (Quan et al., 2025), (Khan et al., 2024), (Ranjithkumar et al., 2021), (Koraag et al., 2022), (Sudarman et al., 2024), (Huang et al., 2024).



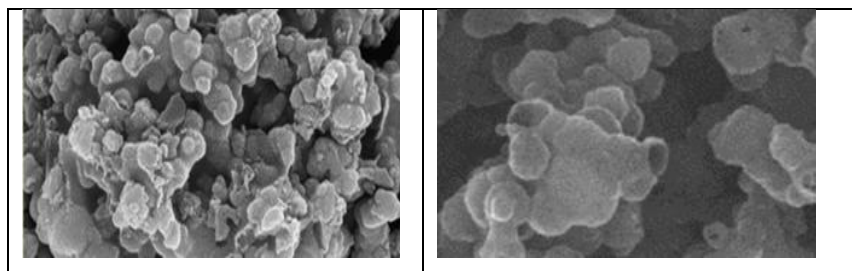
Lampiran 5. 2 Sampel Primer Penelitian

Sumber : Dokumen pribadi data penelitian



Lampiran 5. 3 Sampel Silikon Treated Sekunder Logam

Sumber : (Chen et al., 2020), (Deng et al., 2018),(Luo et al., 2022),(Y.-L. Hou et al., 2021), (Gan et al., 2025), (Jiang & Li, 2022b), (Zhang & Li, 2023), (S.-C. Hou et al., 2018), (Zeng et al., 2022), (Li et al., 2015)



Lampiran 5. 4 Sampel Sekunder Silikon Treated Nitrogen

Sumber : (Mei et al., 2023), (Xie et al., 2021)

DAFTAR PUSTAKA

- Chen, Z., Wang, X., Jian, T., Hou, J., Zhou, J., & Xu, C. (2020). One-step mild fabrication of branch-like multimodal porous Si/Zn composites as high performance anodes for Li-ion batteries. *Solid State Ionics*, 354, 115406. <https://doi.org/10.1016/j.ssi.2020.115406>
- Deng, L., Wu, Z.-Y., Yin, Z.-W., Lu, Y.-Q., Huang, Z.-G., You, J.-H., Li, J.-T., Huang, L., & Sun, S.-G. (2018). High-performance Si Mn/C composite anodes with integrating inactive Mn₄Si₇ alloy for lithium-ion batteries. *Electrochimica Acta*, 260, 830–837. <https://doi.org/10.1016/j.electacta.2017.12.048>
- Gan, X., He, J.-L., He, X.-R., Zhang, K., & Cui, Y.-H. (2025). A N-doped carbon-coated porous Si–Cu composite as a high-performance anode for lithium-ion batteries. *New Journal of Chemistry*, 49(18), 7403–7413. <https://doi.org/10.1039/D5NJ00572H>
- Hou, S.-C., Chen, T.-Y., Wu, Y.-H., Chen, H.-Y., Lin, X.-D., Chen, Y.-Q., Huang, J.-L., & Chang, C.-C. (2018). Mechanochemical synthesis of Si/Cu₃Si-based composite as negative electrode materials for lithium ion battery. *Scientific Reports*, 8(1), 12695. <https://doi.org/10.1038/s41598-018-30703-3>
- Hou, Y.-L., Yang, Y., Meng, W.-J., Lei, B.-Y., Ren, M.-X., Yang, X.-X., Wang, Y.-Q., & Zhao, D.-L. (2021). Core-shell structured Si@Cu nanoparticles encapsulated in carbon cages as high-performance lithium-ion battery anodes. *Journal of Alloys and Compounds*, 874, 159988. <https://doi.org/10.1016/j.jallcom.2021.159988>
- Huang, L., Wang, J., Hu, Y., Chen, C., Cao, L., & Jiang, Q. (2024). Low-cost silicon cutting waste reused as a high-power-density silicon-based anode. *RSC Advances*, 14(47), 34823–34832. <https://doi.org/10.1039/D4RA06203E>
- Jiang, P., & Li, J. (2022a). A porous silicon anode prepared by dealloying a Sr-modified Al–Si eutectic alloy for lithium ion batteries. *RSC Advances*, 12(13), 7892–7897. <https://doi.org/10.1039/D2RA00443G>
- Jiang, P., & Li, J. (2022b). A porous silicon anode prepared by dealloying a Sr-modified Al–Si eutectic alloy for lithium ion batteries. *RSC Advances*, 12(13), 7892–7897. <https://doi.org/10.1039/D2RA00443G>
- Khan, M., Yan, S., Ali, M., Mahmood, F., Zheng, Y., Li, G., Liu, J., Song, X., & Wang, Y. (2024). Innovative Solutions for High-Performance Silicon Anodes in Lithium-Ion Batteries: Overcoming Challenges and Real-World Applications. *Nano-Micro Letters*, 16(1), 179. <https://doi.org/10.1007/s40820-024-01388-3>
- Koraag, P. Y. E., Firdaus, A. M., Hawari, N. H., Refino, A. D., Dempwolf, W., Iskandar, F., Peiner, E., Wasisto, H. S., & Sumboja, A. (2022). Covalently Bonded Ball-Milled Silicon/CNT

- Nanocomposite as Lithium-Ion Battery Anode Material. *Batteries*, 8(10), 165. <https://doi.org/10.3390/batteries8100165>
- Li, Y., Huang, L., Zhang, P., Ren, X., & Deng, L. (2015). Synthesis of Si-Sb-ZnO Composites as High-Performance Anodes for Lithium-ion Batteries. *Nanoscale Research Letters*, 10(1), 414. <https://doi.org/10.1186/s11671-015-1128-4>
- Luo, C., Zhou, X., Ding, J., Yang, J., Zhou, H., Wang, X., & Tang, J. (2022). In-situ migration of Ni induced crystallization to boost the initial coulombic efficiency of nano Si anode for lithium ion batteries. *Composites Communications*, 32, 101157. <https://doi.org/10.1016/j.coco.2022.101157>
- Mei, Y., He, Y., Zhu, H., Ma, Z., Pu, Y., Chen, Z., Li, P., He, L., Wang, W., & Tang, H. (2023). Recent Advances in the Structural Design of Silicon/Carbon Anodes for Lithium Ion Batteries: A Review. *Coatings*, 13(2), 436. <https://doi.org/10.3390/coatings13020436>
- Protopapa, M. L., Buresi, E., Fiore, A., Mirengi, L., Palazzo, B., Schioppa, M., Tagliente, M. A., Valenzano, V., Taurisano, N., & Appetecchi, G. B. (2025). Towards Sustainable Lithium-Ion Batteries: A State-of-the-Art Review on Silicon Anodes, Economics, and Recycling. *Silicon*, 17(17), 3693–3715. <https://doi.org/10.1007/s12633-025-03447-9>
- Quan, L., Su, Q., Lei, H., Zhang, W., Deng, Y., He, J., Lu, Y., Li, Z., Liu, H., Xing, L., & Li, W. (2025). Integrated prelithiation and SEI engineering for high-performance silicon anodes in lithium-ion batteries. *National Science Review*, 12(7). <https://doi.org/10.1093/nsr/nwaf084>
- Ranjithkumar, A., Suresh, C., Eswaran, M., Rajasekar, K., Saravanan, U. M., & Sathees Kumar, T. (2021). Investigation of process parameters of AMEDM on Al/SiC composite using Taguchi approach. *Materials Today: Proceedings*, 45, 2470–2473. <https://doi.org/10.1016/j.matpr.2020.11.027>
- Sudarman, S., Andriyani, Tamrin, & Taufik, M. (2024). Synthesis and application of nano-silicon prepared from rice husk with the hydrothermal method and its use for anode lithium-ion batteries. *Materials Science for Energy Technologies*, 7, 1–8. <https://doi.org/10.1016/j.mset.2023.07.003>
- Wagner, N. P., Tron, A., Tolchard, J. R., Noia, G., & Bellmann, M. P. (2019). Silicon anodes for lithium-ion batteries produced from recovered kerf powders. *Journal of Power Sources*, 414, 486–494. <https://doi.org/10.1016/j.jpowsour.2019.01.035>
- Xiao Zhongliang, Xia Yubo, Wang Cheng, Zhao Tingting, & Yuan Qunxuan. (2024). *Preparation and properties of porous silicon anode coated with nitrogen-doped carbon for lithium-ion battery*. 30, 1–12.

- Xie, Y., He, C.-J., Zhang, J., Hou, Y.-L., Meng, W.-J., & Zhao, D.-L. (2021). Nitrogen-doped carbon caging silicon nanoparticles for high performance lithium-ion battery anodes. *Journal of Alloys and Compounds*, 860, 158487. <https://doi.org/10.1016/j.jallcom.2020.158487>
- Zeng, H., He, Y., & Chamas, M. (2022). Si/Cu composite as anode material for lithium-ion batteries. *Frontiers in Energy Research*, 10. <https://doi.org/10.3389/fenrg.2022.968259>
- Zhang, H., & Li, J. (2023). High-performance SiGe anode materials obtained by dealloying a Sr-modified Al–Si–Ge eutectic precursor. *RSC Advances*, 13(4), 2672–2679. <https://doi.org/10.1039/D2RA07674H>